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Independent · Imperia OS · 2025 – present

Building an AI assistant to do everything its user wants — and the platform that goal requires.

The same infrastructure builds the platform and delivers it to its users.

Two kinds of memory back the development, and both reach the agent the same two ways: pulled into its prompt automatically by embedding match, or fetched on demand through MCP tools.

The first is a knowledge graph that follows the development flow: business use cases come first; tests are written from them; code implements them; and all three are automatically indexed into the graph and embedded.

The second is a thought system: a hierarchy that composes itself, where every new thought is summarized into the layer above, recursively. The same module runs the business agents.

The knowledge map turns re-use into the default. Whatever exists is visible before anything new is written, and Clean Architecture keeps every module portable enough to use anywhere.

Another way of working with memory: state. A stateful agent is a state machine — the algorithm controls the flow, and each step is atomic: an LLM call or a deterministic operation. The algorithm doesn't hallucinate. Stateless agents are the rest: one call, no machine.

Agents are one of three abstractions the system reduces to. Blocks are the second — everything addressable shares one shape and one embedding. Protocols are the third — everything that crosses a boundary. Payment included.

Paid access to the agents is a decorator over the protocol — a block passes the check before reaching the agent. The same slot accepts other decorators — signing, audit, blockchain commitment.

Full-Stack Engineer · Independent · 2022 – 2025

A cross-platform mobile platform on Flutter and AWS. Clean Architecture from the first commit, cloud infrastructure as code in AWS CDK.

Dart and Flutter on the frontend — feature-based modules, event-driven reactive layout, Riverpod for state. AWS Lambda in Go on the backend — DynamoDB with composite keys, S3 for media, Amplify and Cognito for auth.

Blockchain Developer · IBM · 2020 – 2022

Built a full testing environment and CI/CD pipeline for Hyperledger Fabric — feature delivery speed doubled. Refactored the project structure, cutting the codebase by 30%. Led the CouchDB→LevelDB migration that improved data-operation performance by 40%. Delivered a cross-channel encryption solution with zero defects, despite the absence of formal requirements.

Blockchain Developer · Insolar · 2019 – 2020

Designed the data architecture for a blockchain-based wallet, initiating the process and guiding a newly onboarded system architect through the key design decisions.

Blockchain Developer · Sber · 2018 – 2019

A custom CRUD-style abstraction layer over Hyperledger Fabric, self-initiated and built after hours. New-feature implementation dropped from around eight hours to fifteen minutes, simplifying composite-key data operations.

St. Petersburg State University, Mat-Mekh · 2012 – 2017

Mathematics and Mechanics faculty. Arrived from LNMO, the Saint Petersburg laboratory for continuous mathematical education. Olympiad medals along the way — mathematics at ICYS and the City Olympiad, computer science at the Baltic Scientific and Engineering Competition. Worked through university in parallel — Java, Android, Delphi, and embedded C++ for spectrograph instrumentation at a materials research institute.